

CSC 212: Data Structures and Abstractions

15: Binary search trees (part 2)

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Operations: remove

Remove

Approach

- ✓ locate the node to be removed, then apply one of the following cases
- ✓ **case 1**: node is a leaf (no children)
 - delete the node and set the parent's pointer to `nullptr`
- ✓ **case 2**: node has exactly one child
 - bypass the node by updating the parent's pointer to point to the node's only child
 - delete the node
- ✓ **case 3**: node has two children
 - find the inorder **successor** (the smallest node in the right subtree)
 - **alternative**: inorder **predecessor** (the largest node in the left subtree) may be used instead
 - copy the successor's data to the current node
 - recursively delete the successor node (in the right subtree) or predecessor node (in the left subtree)

Time complexity

- ✓ $O(h)$, where h is the height of the tree

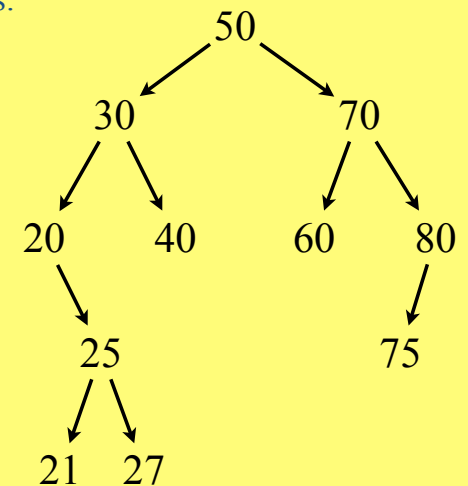
if root is the node to delete, update root pointer after deletion

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Practice

Remove the following keys:

- ✓ 27, 40, 80, 20, 30, 50



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Analysis

Practice

- Starting from an empty BST, insert the following keys in the order given
 - ✓ 20 10 30 5 15 25 35
 - ✓ 10 20 5 15 30 35 25
 - ✓ 5 10 15 20 25 30 35
- ✓ How is the order of insertion related to the shape of the tree?
- ✓ Can you identify worst-case, best-case, and average-case scenarios?

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Average case

Proposition

- ✓ if n distinct keys are inserted in random order into an initially empty BST, the expected number of comparisons for a search is $\sim c \log n$
- this results can be formally justified through probabilistic analysis (beyond the scope of this class)

Implications

- ✓ even without explicit balancing mechanisms, randomly built BSTs provide logarithmic expected time for search, insert, and delete operations

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Practice

- Complete the following table with rates of growth
 - ✓ as a function of the number of nodes

Operation	Best case	Average case	Worst case
Insert			
Remove			
Search			

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Collections

Operation	Description	Sequential (unordered)	Sequential (ordered)	BST
search	search for a key	$O(n)$	$O(\log n)$	$O(h)$
insert	insert a key	$O(n)$	$O(n)$	$O(h)$
delete	delete a key	$O(n)$	$O(n)$	$O(h)$
min/max	find smallest/largest key	$O(n)$	$O(1)$	$O(h)$
floor/ceiling	find predecessor/successor	$O(n)$	$O(\log n)$	$O(h)$
rank	count number of keys less than key	$O(n)$	$O(\log n)$	$O(h)$

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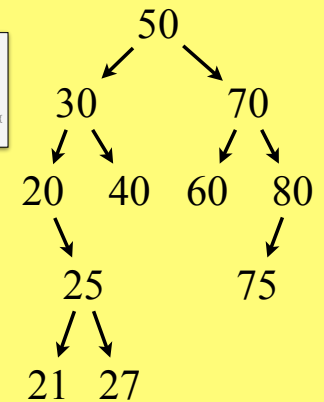
Practice

- Implement a range query operation for a BST, that returns all keys within a specified range $[\min, \max]$, inclusive

✓ trace a call with $[25, 61]$

```
template <typename T>
void BST::range(const T& min, const T& max, std::vector<T>& result) const {
    result.clear();
    range_helper(root, min, max, result);
}

template <typename T>
void BST::range_helper(Node<T>* p, const T& min, const T& max, std::vector<T>& result) const {
    // TODO: Implement this function to collect all keys in the range [min, max]
}
```



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Traversals

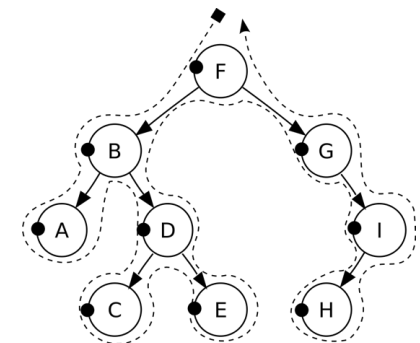
Traversals on (general) trees

Definition

- ✓ a **traversal** is a systematic method for visiting all nodes in a tree exactly once

Types of traversals:

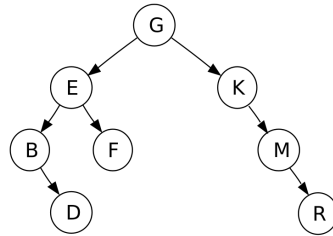
- ✓ **pre-order traversal:** visit root, then recursively traverse all subtrees (left-to-right)
- ✓ **post-order traversal:** recursively traverse all subtrees first (left-to-right), then visit root



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Pre-order traversal

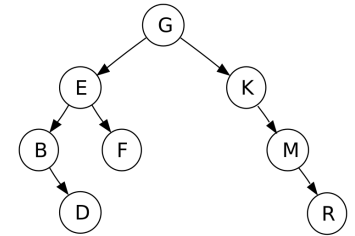
```
algorithm preorder(p) {  
  visit(p)  
  for each child c of p {  
    preorder(c)  
  }  
}
```



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Post-order traversal

```
algorithm postorder(p) {  
  for each child c of p {  
    postorder(c)  
  }  
  visit(p)  
}
```

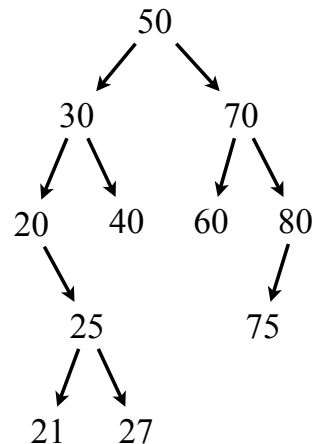


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Preorder traversal on BSTs

- Depth-first traversal that visits the root node first, then recursively traverses the left subtree, followed by the right subtree

- ✓ visit (process) the root node
- ✓ recursively traverse the left subtree
- ✓ recursively traverse the right subtree

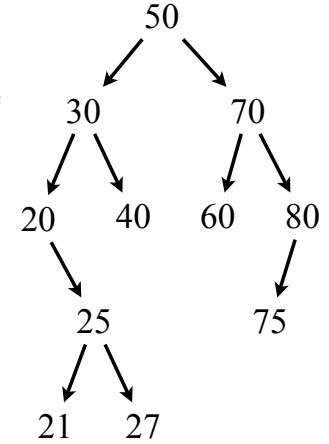


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Postorder traversal on BSTs

- Depth-first traversal that recursively traverses the left subtree, then the right subtree, and finally visits the root node

- ✓ recursively traverse the left subtree
- ✓ recursively traverse the right subtree
- ✓ visit (process) the root node

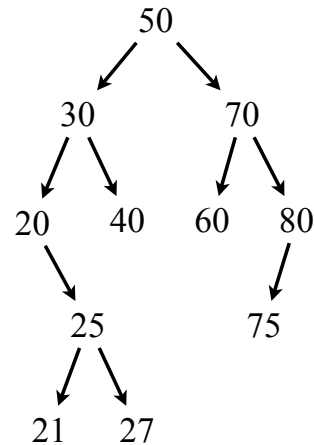


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Inorder traversal on BSTs

- Depth-first traversal that recursively traverses the left subtree, then visits the root node, and finally traverses the right subtree

- ✓ recursively traverse the left subtree
- ✓ visit (process) the root node
- ✓ recursively traverse the right subtree



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Practice

- Which traversal is best for printing all values in sorted order?
- Which traversal is best for deleting all nodes in a tree?
- What is the time complexity of each traversal?

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Practice

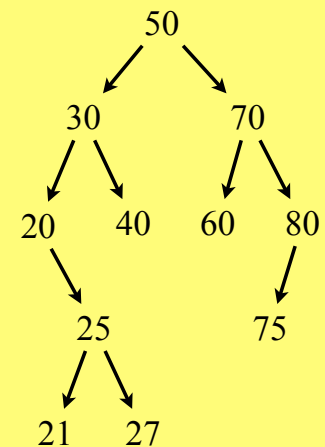
- Write an algorithm that performs preorder traversal on a ternary tree
 - ✓ assume pointers to children are `left`, `mid`, `right`
- Write an algorithm that performs postorder traversal on a k-ary tree with any value of k
 - ✓ assume pointers to children are in an array `children`

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Practice

- Trace the following algorithm and explain what the algorithm prints out

```
algorithm mystery(root) {  
    queue q  
    q.enqueue(root)  
    while not q.isEmpty() {  
        node n = q.dequeue()  
        print(n.value)  
        if n.left  
            q.enqueue(n.left)  
        if n.right  
            q.enqueue(n.right)  
    }  
}
```



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